

11

Support Vector Machines

11.1 Introduction

Fisher's linear discriminant function (LDF) and related classifiers for binary and multiclass learning problems have performed well for many years and for many data sets. Recently, a brand-new learning methodology, *support vector machines (SVMs)*, has emerged (Boser, Guyon, and Vapnik, 1992), which has matched the performance of the LDF and, in many instances, has proved to be superior to it.

Development and implementation of algorithms for SVMs are currently of great interest to theoretical researchers and applied scientists in machine learning, data mining, and bioinformatics. Huge numbers of research articles, tutorials, and textbooks have been published on the topic, and annual workshops, new research journals, courses, and websites are now devoted to the subject. SVMs have been successfully applied to classification problems as diverse as handwritten digit recognition, text categorization, cancer classification using microarray expression data, protein secondary-structure prediction, and cloud classification using satellite-radiance profiles.

SVMs, which are available in both linear and nonlinear versions, involve optimization of a convex loss function under given constraints and so are unaffected by problems of local minima. This gives SVMs quite a strong

competitive advantage over methods such as neural networks and decision trees. SVMs are computed using well-documented, general-purpose, mathematical programming algorithms, and their performance in many situations has been quite remarkable. Even in the face of massive data sets, extremely fast and efficient software is being designed to compute SVMs for classification.

By means of the new technology of *kernel methods*, SVMs have been very successful in building highly nonlinear classifiers. The kernel method enables us to construct linear classifiers in high-dimensional feature spaces that are nonlinearly related to input space and to carry out those computations in input space using very few parameters. SVMs have also been successful in dealing with situations in which there are many more variables than observations.

Although these advantages hold in general, we have to recognize that there will always be applications in which SVMs can get beaten in performance by a hand-crafted classification method.

In this chapter, we describe the linear and nonlinear SVM as solutions of the binary classification problem. The nonlinear SVM incorporates nonlinear transformations of the input vectors and uses the kernel trick to simplify computations. We describe a variety of kernels, including string kernels for text categorization problems. Although the SVM methodology was built specifically for binary classification, we discuss attempts to extend that methodology to multiclass classification. Finally, although the SVM methodology was originally designed to solve classification problems, we discuss how the SVM methodology has been defined for regression situations.

11.2 Linear Support Vector Machines

Assume we have available a learning set of data,

$$\mathcal{L} = \{(\mathbf{x}_i, y_i) : i = 1, 2, \dots, n\}, \quad (11.1)$$

where $\mathbf{x}_i \in \mathbb{R}^r$ and $y_i \in \{-1, +1\}$. The *binary classification problem* is to use $\mathcal{L}$ to construct a function $f : \mathbb{R}^r \rightarrow \mathbb{R}$ so that

$$C(\mathbf{x}) = \text{sign}(f(\mathbf{x})) \quad (11.2)$$

is a classifier. The *separating function* f then classifies each new point $\mathbf{x}$ in a test set $\mathcal{T}$ into one of two classes, Π_+ or Π_- , depending upon whether $C(\mathbf{x})$ is $+1$ (if $f(\mathbf{x}) \geq 0$) or -1 (if $f(\mathbf{x}) < 0$), respectively. The goal is to have f assign all positive points in $\mathcal{T}$ (i.e., those with $y = +1$) to Π_+ and

all negative points in $\mathcal{T}$ ($y = -1$) to Π_- . In practice, we recognize that 100% correct classification may not be possible.

11.2.1 The Linearly Separable Case

First, consider the simplest situation: suppose the positive ($y_i = +1$) and negative ($y_i = -1$) data points from the learning set $\mathcal{L}$ can be separated by a hyperplane,

$$\{\mathbf{x} : f(\mathbf{x}) = \beta_0 + \mathbf{x}^\tau \boldsymbol{\beta} = 0\}, \quad (11.3)$$

where $\boldsymbol{\beta}$ is the *weight vector* with Euclidean norm $\|\boldsymbol{\beta}\|$, and β_0 is the *bias*. (Note: $b = -\beta_0$ is the *threshold*.) If this hyperplane can separate the learning set into the two given classes without error, the hyperplane is termed a *separating hyperplane*. Clearly, there is an infinite number of such separating hyperplanes. How do we determine which one is the best?

Consider any separating hyperplane. Let d_- be the shortest distance from the separating hyperplane to the nearest negative data point, and let d_+ be the shortest distance from the same hyperplane to the nearest positive data point. Then, the *margin* of the separating hyperplane is defined as $d = d_- + d_+$. If, in addition, the distance between the hyperplane and its closest observation is maximized, we say that the hyperplane is an *optimal separating hyperplane* (also known as a *maximal margin classifier*).

If the learning data from the two classes are linearly separable, there exists β_0 and $\boldsymbol{\beta}$ such that

$$\beta_0 + \mathbf{x}_i^\tau \boldsymbol{\beta} \geq +1, \quad \text{if } y_i = +1, \quad (11.4)$$

$$\beta_0 + \mathbf{x}_i^\tau \boldsymbol{\beta} \leq -1, \quad \text{if } y_i = -1. \quad (11.5)$$

If there are data vectors in $\mathcal{L}$ such that equality holds in (11.4), then these data vectors lie on the hyperplane H_{+1} : $(\beta_0 - 1) + \mathbf{x}^\tau \boldsymbol{\beta} = 0$; similarly, if there are data vectors in $\mathcal{L}$ such that equality holds in (11.5), then these data vectors lie on the hyperplane H_{-1} : $(\beta_0 + 1) + \mathbf{x}^\tau \boldsymbol{\beta} = 0$. Points in $\mathcal{L}$ that lie on either one of the hyperplanes H_{-1} or H_{+1} , are said to be *support vectors*. See Figure 11.1. The support vectors typically consist of a small percentage of the total number of sample points.

If $\mathbf{x}_{-1}$ lies on the hyperplane H_{-1} , and if $\mathbf{x}_{+1}$ lies on the hyperplane H_{+1} , then,

$$\beta_0 + \mathbf{x}_{-1}^\tau \boldsymbol{\beta} = -1, \quad \beta_0 + \mathbf{x}_{+1}^\tau \boldsymbol{\beta} = +1. \quad (11.6)$$

The difference of these two equations is $\mathbf{x}_{+1}^\tau \boldsymbol{\beta} - \mathbf{x}_{-1}^\tau \boldsymbol{\beta} = 2$, and their sum is $\beta_0 = -\frac{1}{2}\{\mathbf{x}_{+1}^\tau \boldsymbol{\beta} + \mathbf{x}_{-1}^\tau \boldsymbol{\beta}\}$. The perpendicular distances of the hyperplane $\beta_0 + \mathbf{x}^\tau \boldsymbol{\beta} = 0$ from the points $\mathbf{x}_{-1}$ and $\mathbf{x}_{+1}$ are

$$d_- = \frac{|\beta_0 + \mathbf{x}_{-1}^\tau \boldsymbol{\beta}|}{\|\boldsymbol{\beta}\|} = \frac{1}{\|\boldsymbol{\beta}\|}, \quad d_+ = \frac{|\beta_0 + \mathbf{x}_{+1}^\tau \boldsymbol{\beta}|}{\|\boldsymbol{\beta}\|} = \frac{1}{\|\boldsymbol{\beta}\|}, \quad (11.7)$$

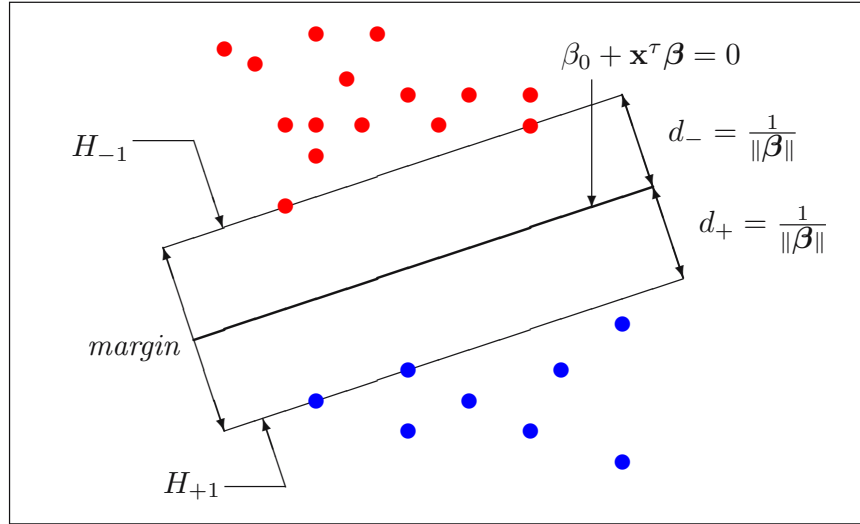


FIGURE 11.1. Support vector machines: the linearly separable case. The red points correspond to data points with $y_i = -1$, and the blue points correspond to data points with $y_i = +1$. The separating hyperplane is the line $\beta_0 + \mathbf{x}^T \beta = 0$. The support vectors are those points lying on the hyperplanes H_{-1} and H_{+1} . The margin of the separating hyperplane is $d = 2 / \|\beta\|$.

respectively (see Exercise 11.1). So, the margin of the separating hyperplane is $d = 2 / \|\beta\|$.

The inequalities (11.4) and (11.5) can be combined into a single set of inequalities,

$$y_i(\beta_0 + \mathbf{x}_i^T \beta) \geq +1, \quad i = 1, 2, \dots, n. \quad (11.8)$$

The quantity $y_i(\beta_0 + \mathbf{x}_i^T \beta)$ is called the *margin of $(\mathbf{x}_i, y_i)$ with respect to the hyperplane (11.3)*, $i = 1, 2, \dots, n$. From (11.6), we see that $\mathbf{x}_i$ is a support vector with respect to the hyperplane (11.3) if its margin equals one; that is, if

$$y_i(\beta_0 + \mathbf{x}_i^T \beta) = 1. \quad (11.9)$$

The support vectors in Figure 11.1 are identified (with circles around them). The empirical distribution of the margins of all the observations in $\mathcal{L}$ is called the *margin distribution of a hyperplane with respect to $\mathcal{L}$* . The minimum of the empirical margin distribution is the *margin of the hyperplane with respect to $\mathcal{L}$* .

The problem is to find the optimal separating hyperplane; namely, find the hyperplane that maximizes the margin, $2 / \|\beta\|$, subject to the conditions (11.8). Equivalently, we wish to find β_0 and β to

$$\text{minimize} \quad \frac{1}{2} \|\beta\|^2, \quad (11.10)$$

$$\text{subject to} \quad y_i(\beta_0 + \mathbf{x}_i^T \beta) \geq 1, \quad i = 1, 2, \dots, n. \quad (11.11)$$

This is a convex optimization problem: minimize a quadratic function subject to linear inequality constraints. Convexity ensures that we have a global minimum without local minima. The resulting optimal separating hyperplane is called the *maximal* (or *hard*) *margin solution*.

We solve this problem using Lagrangian multipliers. Because the constraints are $y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) - 1 \geq 0$, $i = 1, 2, \dots, n$, we multiply the constraints by positive Lagrangian multipliers and subtract each such product from the objective function (11.10) to form the *primal functional*,

$$F_P(\beta_0, \boldsymbol{\beta}, \boldsymbol{\alpha}) = \frac{1}{2} \|\boldsymbol{\beta}\|^2 - \sum_{i=1}^n \alpha_i \{y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) - 1\}, \quad (11.12)$$

where

$$\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^T \geq \mathbf{0} \quad (11.13)$$

is the n -vector of (nonnegative) Lagrangian coefficients. We need to minimize F with respect to the *primal variables* β_0 and $\boldsymbol{\beta}$, and then maximize the resulting minimum- F with respect to the *dual variables* $\boldsymbol{\alpha}$.

The Karush–Kuhn–Tucker conditions give necessary and sufficient conditions for a solution to a constrained optimization problem. For our primal problem, β_0 , $\boldsymbol{\beta}$, and $\boldsymbol{\alpha}$ have to satisfy:

$$\frac{\partial F_P(\beta_0, \boldsymbol{\beta}, \boldsymbol{\alpha})}{\partial \beta_0} = -\sum_{i=1}^n \alpha_i y_i = 0, \quad (11.14)$$

$$\frac{\partial F_P(\beta_0, \boldsymbol{\beta}, \boldsymbol{\alpha})}{\partial \boldsymbol{\beta}} = \boldsymbol{\beta} - \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i = \mathbf{0}, \quad (11.15)$$

$$y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) - 1 \geq 0, \quad (11.16)$$

$$\alpha_i \geq 0, \quad (11.17)$$

$$\alpha_i \{y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) - 1\} = 0, \quad (11.18)$$

for $i = 1, 2, \dots, n$. The condition (11.18) is known as the *Karush–Kuhn–Tucker complementarity condition*.

Solving equations (11.14) and (11.15) yields

$$\sum_{i=1}^n \alpha_i y_i = 0, \quad (11.19)$$

$$\boldsymbol{\beta}^* = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i. \quad (11.20)$$

Substituting (11.19) and (11.20) into (11.12) yields the minimum value of $F_P(\beta_0, \boldsymbol{\beta}, \boldsymbol{\alpha})$, namely,

$$F_D(\boldsymbol{\alpha}) = \frac{1}{2} \|\boldsymbol{\beta}^*\|^2 - \sum_{i=1}^n \alpha_i \{y_i(\beta_0^* + \mathbf{x}_i^T \boldsymbol{\beta}^*) - 1\}$$

$$\begin{aligned}
&= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^\tau \mathbf{x}_j) - \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^\tau \mathbf{x}_i) + \sum_{i=1}^n \alpha_i \\
&= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^\tau \mathbf{x}_j), \tag{11.21}
\end{aligned}$$

where we used (11.18) in the second line. Note that the primal variables have been removed from the problem. The expression (11.21) is usually referred to as the *dual functional* of the optimization problem.

We next find the Lagrangian multipliers $\boldsymbol{\alpha}$ by maximizing the dual functional (11.21) subject to the constraints (11.17) and (11.19). The constrained maximization problem (the “Wolfe dual”) can be written in matrix notation as follows. Find $\boldsymbol{\alpha}$ to

$$\text{maximize } F_D(\boldsymbol{\alpha}) = \mathbf{1}_n^\tau \boldsymbol{\alpha} - \frac{1}{2} \boldsymbol{\alpha}^\tau \mathbf{H} \boldsymbol{\alpha} \tag{11.22}$$

$$\text{subject to } \boldsymbol{\alpha} \geq \mathbf{0}, \boldsymbol{\alpha}^\tau \mathbf{y} = \mathbf{0}, \tag{11.23}$$

where $\mathbf{y} = (y_1, \dots, y_n)^\tau$ and $\mathbf{H} = (H_{ij})$ is a square $(n \times n)$ -matrix with $H_{ij} = y_i y_j (\mathbf{x}_i^\tau \mathbf{x}_j)$. If $\hat{\boldsymbol{\alpha}}$ solves this optimization problem, then

$$\hat{\boldsymbol{\beta}} = \sum_{i=1}^n \hat{\alpha}_i y_i \mathbf{x}_i \tag{11.24}$$

yields the optimal weight vector. If $\hat{\alpha}_i > 0$, then, from (11.18), $y_i(\beta_0^* + \mathbf{x}_i^\tau \boldsymbol{\beta}^*) = 1$, and so $\mathbf{x}_i$ is a support vector; for all observations that are not support vectors, $\hat{\alpha}_i = 0$. Let $sv \subset \{1, 2, \dots, n\}$ be the subset of indices that identify the support vectors (and also the nonzero Lagrangian multipliers). Then, the optimal $\boldsymbol{\beta}$ is given by (11.24), where the sum is taken only over the support vectors; that is,

$$\hat{\boldsymbol{\beta}} = \sum_{i \in sv} \hat{\alpha}_i y_i \mathbf{x}_i. \tag{11.25}$$

In other words, $\hat{\boldsymbol{\beta}}$ is a linear function only of the support vectors $\{\mathbf{x}_i, i \in sv\}$. In most applications, the number of support vectors will be small relative to the size of $\mathcal{L}$, yielding a *sparse* solution. In this case, the support vectors carry all the information necessary to determine the optimal hyperplane.

The primal and dual optimization problems yield the same solution, although the dual problem is simpler to compute and, as we shall see, is simpler to generalize to nonlinear classifiers. Finding the solution involves standard convex quadratic-programming methods, and so any local minimum also turns out to be a global minimum.

Although the optimal bias $\hat{\beta}_0$ is not determined explicitly by the optimization solution, we can estimate it by solving (11.18) for each support

vector and then averaging the results. In other words, the estimated bias of the optimal hyperplane is given by

$$\hat{\beta}_0 = \frac{1}{|sv|} \sum_{i \in sv} \left(\frac{1 - y_i \mathbf{x}_i^\tau \hat{\boldsymbol{\beta}}}{y_i} \right), \quad (11.26)$$

where $|sv|$ is the number of support vectors in $\mathcal{L}$.

It follows that the optimal hyperplane can be written as

$$\begin{aligned} \hat{f}(\mathbf{x}) &= \hat{\beta}_0 + \mathbf{x}^\tau \hat{\boldsymbol{\beta}} \\ &= \hat{\beta}_0 + \sum_{i \in sv} \hat{\alpha}_i y_i (\mathbf{x}^\tau \mathbf{x}_i). \end{aligned} \quad (11.27)$$

Clearly, only support vectors are relevant in computing the optimal separating hyperplane; observations that are not support vectors play no role in determining the hyperplane and are, thus, irrelevant to solving the optimization problem. The *classification rule* is given by

$$C(\mathbf{x}) = \text{sign}\{\hat{f}(\mathbf{x})\}. \quad (11.28)$$

If $j \in sv$, then, from (11.27),

$$y_j \hat{f}(\mathbf{x}_j) = y_j \hat{\beta}_0 + \sum_{i \in sv} \hat{\alpha}_i y_i y_j (\mathbf{x}_j^\tau \mathbf{x}_i) = 1. \quad (11.29)$$

Hence, the squared-norm of the weight vector $\hat{\boldsymbol{\beta}}$ of the optimal hyperplane is

$$\begin{aligned} \|\hat{\boldsymbol{\beta}}\|^2 &= \sum_{i \in sv} \sum_{j \in sv} \hat{\alpha}_i \hat{\alpha}_j y_i y_j (\mathbf{x}_i^\tau \mathbf{x}_j) \\ &= \sum_{j \in sv} \hat{\alpha}_j y_j \sum_{i \in sv} \hat{\alpha}_i y_i (\mathbf{x}_i^\tau \mathbf{x}_j) \\ &= \sum_{j \in sv} \hat{\alpha}_j (1 - y_j \hat{\beta}_0) \\ &= \sum_{j \in sv} \hat{\alpha}_j. \end{aligned} \quad (11.30)$$

The third line used (11.29) and the fourth line used (11.19). It follows from (11.30) that the optimal hyperplane has maximum margin $2/\|\hat{\boldsymbol{\beta}}\|$, where

$$\frac{1}{\|\hat{\boldsymbol{\beta}}\|} = \left(\sum_{j \in sv} \hat{\alpha}_j \right)^{-1/2}. \quad (11.31)$$

11.2.2 The Linearly Nonseparable Case

In real applications, it is unlikely that there will be such a clear linear separation between data drawn from two classes. More likely, there will be some overlap. We can generally expect some data from one class to infiltrate the region of space perceived to belong to the other class, and vice versa. The overlap will cause problems for any classification rule, and, depending upon the extent of the overlap, we should expect that some of the overlapping points will be misclassified.

The *nonseparable case* occurs if either the two classes are separable, but not linearly so, or that no clear separability exists between the two classes, linearly or nonlinearly. One reason for overlapping classes is the high noise level (i.e., large variances) of one or both classes. As a result, one or more of the constraints will be violated.

The way we cope with overlapping data is to create a more flexible formulation of the problem, which leads to a *soft-margin solution*. To do this, we introduce the concept of a nonnegative *slack variable*, ξ_i , for each observation, $(\mathbf{x}_i, y_i)$, in $\mathcal{L}$, $i = 1, 2, \dots, n$. See Figure 11.2 for a two-dimensional example. Let

$$\boldsymbol{\xi} = (\xi_1, \dots, \xi_n)^T \geq \mathbf{0}. \quad (11.32)$$

The constraints (11.11) now become $y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) + \xi_i \geq 1$ for $i = 1, 2, \dots, n$. Data points that obey these constraints have $\xi_i = 0$. The classifier now has to find the optimal hyperplane that controls both the margin, $2/\|\boldsymbol{\beta}\|$, and some computationally simple function of the slack variables, such as

$$g_\sigma(\boldsymbol{\xi}) = \sum_{i=1}^n \xi_i^\sigma, \quad (11.33)$$

subject to certain constraints. The usual values of σ are 1 (“1-norm”) or 2 (“2-norm”). Here, we discuss the case of $\sigma = 1$; for $\sigma = 2$, see Exercise 11.2.

The *1-norm soft-margin optimization problem* is to find β_0 , $\boldsymbol{\beta}$, and $\boldsymbol{\xi}$ to

$$\text{minimize} \quad \frac{1}{2} \|\boldsymbol{\beta}\|^2 + C \sum_{i=1}^n \xi_i, \quad (11.34)$$

$$\text{subject to} \quad \xi_i \geq 0, \quad y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) \geq 1 - \xi_i, \quad i = 1, 2, \dots, n, \quad (11.35)$$

where $C > 0$ is a *regularization parameter*. C takes the form of a tuning constant that controls the size of the slack variables and balances the two terms in the minimizing function.

Form the primal functional, $F_P = F_P(\beta_0, \boldsymbol{\beta}, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\eta})$, where

$$F_P = \frac{1}{2} \|\boldsymbol{\beta}\|^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i \{y_i(\beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}) - (1 - \xi_i)\} - \sum_{i=1}^n \eta_i \xi_i, \quad (11.36)$$

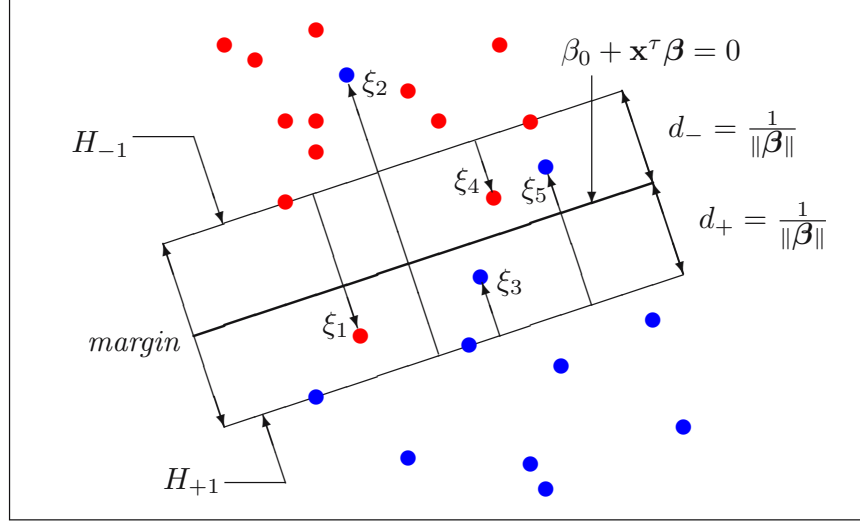


FIGURE 11.2. Support vector machines: the nonlinearly separable case. The red points correspond to data points with $y_i = -1$, and the blue points correspond to data points with $y_i = +1$. The separating hyperplane is the line $\beta_0 + \mathbf{x}^T \beta = 0$. The support vectors are those circled points lying on the hyperplanes H_{-1} and H_{+1} . The slack variables ξ_1 and ξ_4 are associated with the red points that violate the constraint of hyperplane H_{-1} , and points marked by ξ_2, ξ_3 , and ξ_5 are associated with the blue points that violate the constraint of hyperplane H_{+1} . Points that satisfy the constraints of the appropriate hyperplane have $\xi_i = 0$.

with $\alpha = (\alpha_1, \dots, \alpha_n)^T \geq \mathbf{0}$ and $\eta = (\eta_1, \dots, \eta_n)^T \geq \mathbf{0}$. Fix α and η , and differentiate F_P with respect to β_0 , β , and ξ :

$$\frac{\partial F_P}{\partial \beta_0} = - \sum_{i=1}^n \alpha_i y_i, \quad (11.37)$$

$$\frac{\partial F_P}{\partial \beta} = \beta - \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i, \quad (11.38)$$

$$\frac{\partial F_P}{\partial \xi_i} = C - \alpha_i - \eta_i, \quad i = 1, 2, \dots, n. \quad (11.39)$$

Setting these derivatives equal to zero and solving yields

$$\sum_{i=1}^n \alpha_i y_i = 0, \quad \beta^* = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i, \quad \alpha_i = C - \eta_i. \quad (11.40)$$

Substituting (11.37) into (11.33) gives the dual functional,

$$F_D(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i^T \mathbf{x}_j), \quad (11.41)$$

which, remarkably, is the same as (11.18) for the linearly separable case. From the constraints $C - \alpha_i - \eta_i = 0$ and $\eta_i \geq 0$, we have that $0 \leq \alpha_i \leq C$. In addition, we have the Karush–Kuhn–Tucker conditions:

$$y_i(\beta_0 + \mathbf{x}_i^\tau \boldsymbol{\beta}) - (1 - \xi_i) \geq 0 \quad (11.42)$$

$$\xi_i \geq 0, \quad (11.43)$$

$$\alpha_i \geq 0, \quad (11.44)$$

$$\eta_i \geq 0, \quad (11.45)$$

$$\alpha_i \{y_i(\beta_0 + \mathbf{x}_i^\tau \boldsymbol{\beta}) - (1 - \xi_i)\} = 0, \quad (11.46)$$

$$\xi_i(\alpha_i - C) = 0, \quad (11.47)$$

for $i = 1, 2, \dots, n$. From (11.47), a slack variable, ξ_i , can be nonzero only if $\alpha_i = C$. The Karush–Kuhn–Tucker complementarity conditions, (11.46) and (11.47), can be used to find the optimal bias β_0 .

We can write the dual maximization problem in matrix notation as follows. Find $\boldsymbol{\alpha}$ to

$$\text{maximize } F_D(\boldsymbol{\alpha}) = \mathbf{1}_n^\tau \boldsymbol{\alpha} - \frac{1}{2} \boldsymbol{\alpha}^\tau \mathbf{H} \boldsymbol{\alpha} \quad (11.48)$$

$$\text{subject to } \boldsymbol{\alpha}^\tau \mathbf{y} = 0, \quad \mathbf{0} \leq \boldsymbol{\alpha} \leq C \mathbf{1}_n. \quad (11.49)$$

The only difference between this optimization problem and that for the linearly separable case, (11.22) and (11.23), is that, here, the Lagrangian coefficients α_i , $i = 1, 2, \dots, n$, are each bounded above by C ; this upper bound restricts the influence of each observation in determining the solution. This type of constraint is referred to as a *box constraint* because $\boldsymbol{\alpha}$ is constrained by the box of side C in the positive orthant. From (11.49), we see that the *feasible region* for the solution to this convex optimization problem is the intersection of the hyperplane $\boldsymbol{\alpha}^\tau \mathbf{y} = 0$ with the box constraint $\mathbf{0} \leq \boldsymbol{\alpha} \leq C \mathbf{1}_n$. If $C = \infty$, then the problem reduces to the hard-margin separable case.

If $\hat{\boldsymbol{\alpha}}$ solves this optimization problem, then,

$$\hat{\boldsymbol{\beta}} = \sum_{i \in sv} \hat{\alpha}_i y_i \mathbf{x}_i \quad (11.50)$$

yields the optimal weight vector, where the set sv of support vectors contains those observations in $\mathcal{L}$ which satisfy the constraint (11.42).

11.3 Nonlinear Support Vector Machines

So far, we have discussed methods for constructing a linear SVM classifier. But what if a linear classifier is not appropriate for the data set in